# Synapses

## Modelling presynaptic and postsynaptic mechanisms

Computational Neuroscience
University of Bristol
Adapted in part from B Mizusaki's slides
M Rule

**Learning outcomes:**

- Understand how voltage triggers release of chemical neurotransmitters at the presynapse
- Work with ODE models (mass-action chemical kinetics) of synaptic release
- Understand abstract representations of synapses
- Write down a simple model for a synaptic input

*Start by drawing pictures to remind ourselves of the steps in synaptic transmission.*

*Presynaptic*

1. Action potential arrives
2. Voltage gated calcium channels open
3. $Ca^{2+}$ enters the presynaptic cell
4. $Ca^{2+}$ triggers vesicle fusion

Do not let the lecturer leave this slide without drawing a picture!

# *Quantal release*

Readily-releasable pool

- A certain amount of vesicles loaded with neurotransmitter are docked near the presynaptic membrane, and bound to associated proteins that enable rapid, calcium-triggered release.

Spike arrival at axon is all-or-nothing

- Depolarisation and calcium influx triggers excocytosis of this pool

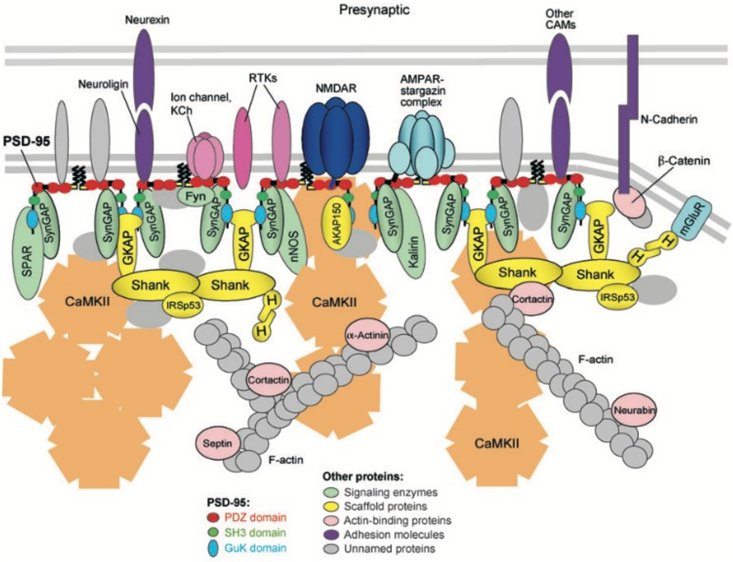
Do not let the lecturer leave this slide without drawing a picture!

***Postsynaptic***

1. Neurotransmitter binds to receptors on postsynaptic membrane
2. Ionotropic receptors open, allowing ionic currents
   Metabotropic receptors trigger signalling cascades that modify ionic conductances
3. Ionic currents change dendritic voltage

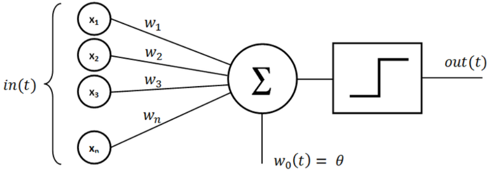
Do not let the lecturer leave this slide without drawing a picture!

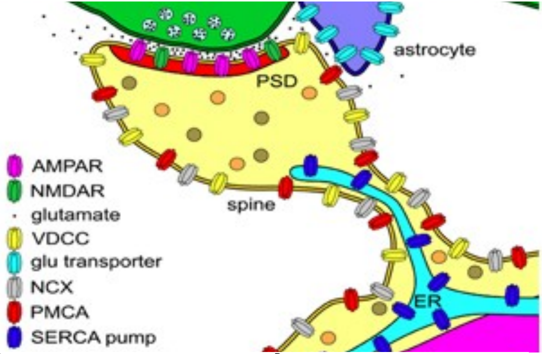
*How should we model this?*

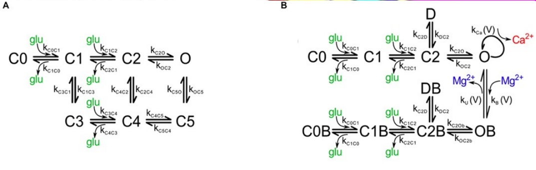



*Sheng & Hoogenraad, Annu Rev Neurosci (2007)*

*Many possible levels of modelling abstraction*




*AMPAR*

*NMDAR-GluN2A/2B*

# *What goes in a model of a synapse?*

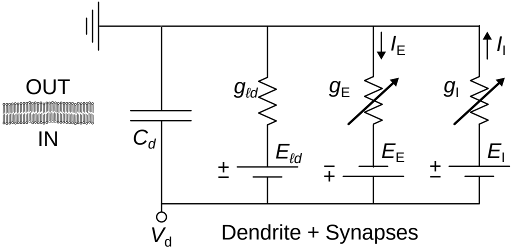
Synapses are unreliable (successful vesicle fusion, release is probabilistic)

Weighted transformation of input amplitudes, non-linear;

Excitatory or inhibitory (depending on presynaptic neuron);

# A common phenomenological model

Time-dependent conductance $\bar{g}_s s(t)$ in series with a battery (voltage $E_s$)




$$I_s(t) = \bar{g}_s s(t) [E_s - V(t)]$$

$\bar{g}_s$: ***maximal conductance*** $\sim \#$ receptors in synapse $\times$ conductance when open

$E_s$: Reversal potential for ionotropic receptor

- $\sim 0$ for excitatory synapses ($K^+$, $Na^+$)
- $\sim V_{\mathrm{rest}}$ for inhibitory synapses ($Cl^-$)

***But what is*** $s(t)$ ***??***

***But what is** $s(t)$ **??***

## *A detailed approach: Mass-action chemical kinetics*

much simpler than molecular simultion

often still too complicated

# *Mass-action chemical kinetics: Ionotropic receptor*

$$\text{neurotransmitter} + \text{receptor} \underset{\text{dissociate}}{\overset{\text{bind}}{\rightleftharpoons}} \text{receptor–neurotransmitter complex}$$

$$A + R_{\text{closed}} \underset{\beta}{\overset{\alpha}{\rightleftharpoons}} R_{\text{open}}$$

Fraction open $s$ for fixed $A$ ($A \gg R_{\text{tot}}$)?

$$R_{\text{tot}} := R_{\text{open}} + R_{\text{closed}} \qquad \text{total postsynaptic receptor}$$

$$s := R_{\text{open}}/R_{\text{tot}} \qquad \text{fraction receptor bound}$$

$$\rho_{\text{on}} = \alpha A R_{\text{tot}}(1-s) \qquad \text{Reaction rates}$$

$$\rho_{\text{off}} = \beta R_{\text{tot}} s$$

$$\dot{s} = -\beta s + \alpha A(1-s) \qquad \text{ODE for fraction bound}$$

$$s_\infty(A) = \left[1 + \frac{\beta}{\alpha A}\right]^{-1} = \sigma\left(\ln A - \ln \frac{\beta}{\alpha}\right) \qquad \text{Equilibrium } (\dot{s} = 0)$$

$\sigma(q) = [1 + e^{-q}]^{-1}$ is the logistic sigmoid

this form will re-appear later in different contexts

# *Mass-action chemical reaction kinetics: Ionotropic receptor*

Model 1: Binding (ignoring clearance), solvable

$$\dot{s} = -\beta s + \alpha A(1 - s) \quad \text{ODE for fraction bound}$$
$$\dot{A} = -R_{\text{tot}}\dot{s}$$

Model 2: $A$ added from presynapse ($u(t)$), cleared via reuptake (rate $\gamma$)

$$\dot{A} = -\gamma A + u$$

Model 1+2: Even this is too much to solve

$$\dot{s} = -\beta s + \alpha A(1 - s)$$
$$\dot{A} = -R_{\text{tot}}\dot{s} - \gamma A + u$$

***We just need:***

- Post-synaptic conductance goes up, then down again, whenever the presynaptic axonal terminal spikes

For presynaptic spike times $\mathcal{T}$:

$$s(t) = \sum_{t_s \in \mathcal{T}} h_s(t - t_s)$$

Single exponential

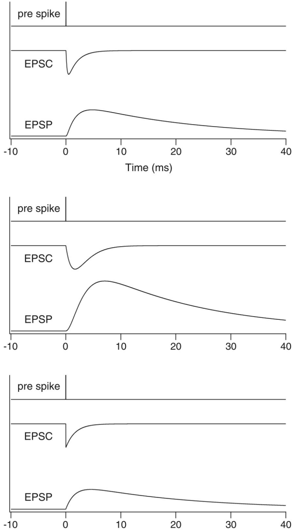
$$h_s(t) = e^{-t/\tau_s}$$

Alpha function

$$h_s(t) = t e^{-t/\tau_s}$$

Difference of exponentials

$$h_s(t) = e^{-t/\tau_{\mathrm{decay}}} - e^{-t/\tau_{\mathrm{rise}}}$$




*Roth & van Rossum*

The lecturer needs to remark on the linear ODE forms for the models on the previous page!

The lecturer may need to briefly discuss convolution here.

# *Summary*

Real synapses are complicated molecular machines

We can model them at multiple levels of granularity, as appropriate for the task at hand

A simple way of modelling a PSP on the postsynaptic membrane potential is with the addition of a single impulse to the current.

# Modelling Scales

… Quantum Chemistry, Molecular dynamics

Physiological, Quantitative

Biological Realism, Data needed to identify parameters

**Molecules**
Gillespie, Master Equation

**Concentrations**
Mass-Action Kinetics

**Conductance Models**
Hodgkin–Huxley

**Spiking Models**
Leaky Integrate and Fire

**Rate Neurons**
Neural Mass/Field Models

**Poisson Neurons**
Generalized Linear Models

**Binary Neurons**
McCulloch–Pitts, Hopfield, Perceptron

Computational Efficiency, Mathematical Tractability

Phenomenological, Qualitative

Cognitive Neuroscience, Psychology …

*end*

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| | |
|---|---|
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